

## Server Requirements

| #  | Component                  | Specification                                                                   |
|----|----------------------------|---------------------------------------------------------------------------------|
| 1  | CPU                        | 8 vCPU / ~8 physical cores (single socket)                                      |
| 2  | RAM                        | 64 GB (optionally 96–128 GB)                                                    |
| 3  | Disk - Log (L:)            | 1× SSD/NVMe, 200–300 GB, RAID1 optional                                         |
| 4  | Disk - Data (D:)           | 2× SSD/NVMe RAID1 or 4× SSD RAID10, 1–2 TB usable                               |
| 5  | Disk - tempdb (T:)         | 1× SSD/NVMe, 200–300 GB, dedicated if possible                                  |
| 6  | Disk - Backup/Staging (B:) | 2–4 TB (SAS HDD + SSD cache acceptable) + external NAS/S3                       |
| 7  | Network                    | 1×10 GbE (1 GbE works; add 2nd NIC if backup/replica traffic)                   |
| 8  | Power/Chassis              | Dual PSU (if physical server)                                                   |
| 9  | OS                         | Windows Server 2022 (High Performance power plan)                               |
| 10 | DB                         | SQL Server 2022 Standard (with Columnstore support)                             |
| 11 | SQL Settings               | Max Memory ~48 GB; MAXDOP=8; Cost Threshold=30–40; tempdb=4 files               |
| 12 | Maintenance                | Daily full or weekly full + daily diff; log backups 15–30 min                   |
| 13 | VMware Notes               | PVSCSI controllers; VMXNET3; Thick Eager Zeroed; High Performance power profile |